

Q1

The set $\{1, 2, \dots, p-1\}$ form a multiplicative group modulo p . Multiplying each element by $a \pmod{p}$ just permutes the set

So,

$$a \cdot 1 \cdot 2 \cdot \dots \cdot (p-1) \equiv (1 \cdot 2 \cdot \dots \cdot (p-1)) \pmod{p}$$

Cancel common terms

$$a^{p-1} \equiv 1 \pmod{p}$$

Example:

$$a=7, p=13$$

$$7^{12} \pmod{13} = 1$$

Q-2

Euler's Totient function $\phi(n)$

$$\phi(25) = 25 \cdot (1 - \frac{1}{5}) = 20$$

$$\phi(45) = 45 \cdot (1 - \frac{1}{3}) \cdot (1 - \frac{1}{5}) = 24$$

$$\phi(100) = 100 \cdot (1 - \frac{1}{2}) \cdot (1 - \frac{1}{5}) = 40$$

17-21058

proof of Euler's Theorem:

if $\gcd(a, n) = 1$ then

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Q-3

Chinese Remainder Theorem

$$\text{Given, } x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{4}$$

$$x \equiv 1 \pmod{5}$$

$$\text{let, } N = 60$$

using CRT formula, we get.

$$x \equiv 11 \pmod{60}$$

verified:

$$11 \pmod{3} \equiv 2$$

$$11 \pmod{4} \equiv 3$$

$$11 \pmod{5} \equiv 1$$

17-2-1058

Q-9] Is 561 a Carmichael Number?

$$561 = 3 \times 11 \times 17 \quad (\text{prime, Distinct})$$

Check $a^{561} \equiv 1 \pmod{561}$ for

$$\gcd(a, 561) = 1$$

561 passes Fermat's test for all such

Hence, 561 is a Carmichael Number.

Q5N5

primitive Root mod 17

Try small integers. Test $g^k \pmod{17}$ of k

+0 = 16,

$$g = 3$$

is a primitive root mod 17

QSN-6 Discrete Logarithm:-

Finding x such that,

$$3^x \equiv 13 \pmod{17}$$

Ans

$$3^1 \rightarrow 3$$

$$3^2 \rightarrow 9$$

$$3^3 \rightarrow 10$$

$$3^4 \rightarrow 13$$

So here $x = 4$.

QSN-7 Discrete log in Diffie-Hellman

DH uses:

public base g , prime p

Each party computes $A = g^a \pmod{p}$

$$B = g^b \pmod{p}$$

shared key: $B^a = A^b \pmod{p}$

Security relies on hardness of Discrete log problem

Q5N-8 Cipher comparison

Cipher	Mechanism	Keyspace	Frequency Analysis
Substitution	Replace each letter	$26!$	Vulnerable
Transposition	Reorder letters	Depend on block size	Medium
Playfair	5x5 digraph grid	$25!$	Harder

plain text: ~~HELLO~~ HELLO

• Substitution (shift) 3 = KHOOR

• Transposition (reverse) : OLLEH

• playfair: custom grid $\rightarrow$ pairs $\rightarrow$ 1H $\rightarrow$ TL

EL $\rightarrow$ KP, LO $\rightarrow$ QR

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QSN-9

A affine cipher $E(x) = (5x+8) \bmod 26$

(a) "DEPT OF ICT MBSTU"

Remove spaces, convert letters to Numbers
(A=0... Z=25)

Encrypt using $E(x) = (5x+8) \bmod 26$

$$D=3 \rightarrow (5 \times 3 + 8) = 23 \rightarrow 2$$

Final cipher text: W X C M Z - W N A E X 2 P I X A

(b) Find modular inverse of $a=5 \bmod 26$

it is 21

$$D(y) = 21(y-8) \bmod 26$$

SN-10

Custom cipher (substitution + permutation)

Design:

substitution : Caesar shift by 3

permutation : Reverse block of 4 letters

Encryption process.

shift letters : HELLO $\rightarrow$ KHOOR

block size 4 : KHOOR

Reverse : OORHK

Decryption process.

1. Reverse the block

2. shift back by 3